SP01 · Longitudinal Strength Calculation

Group 5 · 12 November 2026 · T. Santos, L. Batista

Show that the hull girder carries the still-water and wave bending moments. One calculation note goes in D9, with the drawing listed. The items are in the order of the calculation.

Required documents

- STU-SP01.1 Lightship weight and deadweight distribution along the length.
- STU-SP01.2 Four critical loading conditions (subject to justification and/or definition with the instructor).
- STU-SP01.3 Still-water and wave bending moments and shear forces for each loading condition.
- STU-SP01.4 Allowable stresses, in bending and in shear.
- STU-SP01.5 Preliminary scantlings of the midship cross-section.
- STU-SP01.6 Section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.7 Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them.
- DWG-SP01.1 Midship cross-section.

Checklist

ID	Title	Deliverable	Status
STU-SP01.1	Lightship and deadweight distribution	D9 · SP01	
STU-SP01.2	Four critical loading conditions	D9 · SP01	
STU-SP01.3	Still-water and wave bending moments and shear forces	D9 · SP01	
STU-SP01.4	Allowable stresses, bending and shear	D9 · SP01	
STU-SP01.5	Preliminary scantlings of the midship cross-section	D9 · SP01	
STU-SP01.6	Section modulus, moment of inertia and neutral axis	D9 · SP01	
STU-SP01.7	Allowable still-water bending moment and shear force vs loading conditions	D9 · SP01	
DWG-SP01.1	Midship cross-section	D9 · SP01	